Product rule

1 To differentiate $y = x^6\left(x^4+4\right)$ using the product rule, let $u = x^6$ and $v = x^4+4$, then:

a Find u'.

b Find v'.

c Hence, find $\frac{dy}{dx}$.

2 To differentiate $y = \left(x^7+5\right)\left(x^6+6\right)$ using the product rule, let $u = x^7+5$ and $v = x^6+6$, then:

a Find $\frac{du}{dx}$.

b Find $\frac{dv}{dx}$.

c Hence, find $\frac{dy}{dx}$.

3 Consider the function $y = x^3\left(x^2+9\right)$.

a Differentiate y by first expanding the brackets.

b Differentiate y using the product rule, letting $u = x^3$ and $v = x^2+9$.

4 Differentiate the following functions:

a $f(x) = (3x-2)(4x-5)$

b $g(t) = \left(2t^3-3\right)(3-t)$

c $g(y) = \left(7y^4-y^2\right)\left(y^2-5\right)$

d $f(x) = \left(x^{\frac{4}{3}}+6\sqrt{x}\right)(6x+3)$

e $f(x) = x\sqrt{3-x}$

f $f(x) = \sqrt[3]{x^2}\left(2x-x^2\right)$

g $f(x) = x^{\frac{1}{3}}(1-x)^{\frac{2}{3}}$

h $y = (2+\sqrt{x})\left(6-x^2\right)$

i $y = \left(1+\frac{1}{x}\right)\left(3+x-x^2\right)$

j $y = x^3(5x+3)^7$

k $y = 6x^5\left(x^2+3\right)^3$

l $y = 3x\left(x^2+x+1\right)^9$

m $y = (8x-9)^5(5x+7)^7$

n $y = (3x+2)\sqrt{5+4x}$

o $y = 8x(5+8x)^{\frac{7}{4}}-3$

p $y = 8x^5\sqrt{8x+3}$

q $y = 6x\sqrt{x+1}$

r $y = -4x\sqrt{1-2x}$

5 Consider the function $y = (4x-3)(5x-2)$.

a Differentiate y.

b Hence, differentiate $f(x) = x^3(4x-3)(5x-2)$.

6 For each of the following functions:

i Identify possible factors u and v for the function.

ii Differentiate the function. Give your answer in factorised form.

iii State the values of x for which the derivative is zero.

a $y = x\left(x-8\right)^4$

b $y = x^3\left(x+3\right)^4$

c $y = \left(x+2\right)\left(x+5\right)^6$

7 The derivative of $f\left(x\right) = \left(3x^n+4\right)\left(5x^2-2x\right)$ is of degree 5. Find the value of n.

Gradients, tangents and normals

8 Consider the function $f\left(x\right) = \left(x^2-3x\right)\left(2x-5\right)$.

a Find $f\left(3\right)$.

b Find $f'\left(0\right)$.

c Find $f'\left(-3\right)$.

9 Consider the function $g\left(x\right) = x^3 f\left(x\right)$, where $f\left(x\right)$ is a function of x. Given that $f\left(3\right) = 1$ and $f'\left(3\right) = -3$, find $g'\left(3\right)$.

10 Consider the function $f\left(x\right) = \left(x+1\right)\left(x+3\right)^3$.

a Find $f'\left(x\right)$ in factorised form.

b Find the equation of the tangent at $\left(-1, 0\right)$.

c Find the equation of the normal at $\left(-1, 0\right)$.

11 Find the values of x such that the gradient of the tangent to the curve $y = 2x\left(x+3\right)^2$ is equal to 14.

12 Find the gradient of the tangent to the curve $y = x\sqrt{2x+5}$ at the point where $x = 2$.